Main Effects

and

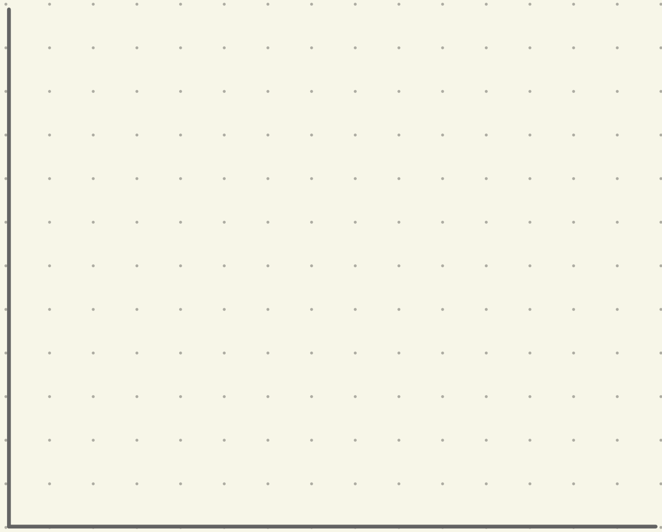
Interactions

Observation from last time

In the linear model,

$$Y_i = \mu + \alpha_{j(i)} + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$



The F-statistic

"The variability of the group averages

relative to the

average variability within each group."

$$\frac{A}{B} = \frac{MS_B}{MS_W} = \frac{\frac{1}{J-1}\sum_{j=1}^{J} n_j\left(\hat{\bar{Y}}_j - \hat{\bar{Y}}\right)^2}{\frac{1}{n-J}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_{j(i)}\right)^2}$$

[Slides]

Estimates